Supplementary Material for Load-Bearing but Redundant: Rethinking Physics Injection in Shared-Latent World Models

Anonymous submission

A Training and Evaluation Details

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder controls.

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3}, bf16, gradient clip 1.0, 20 epochs; the from-scratch cells of the pretraining-regime 2×2 train longer and at a lower learning rate so their baselines converge cleanly—60 epochs on uniform motion and collision, 120 on parabola, which needed the extra budget—with injection on and off sharing the identical budget within each domain, so every Δ in Table 2 of the main paper is a within-domain comparison. Batch size 64 for free rollout ($H{=}8$), 128 for teacher forcing ($H{=}1$), 48 for $H{\geq}16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Both arms of the pretraining-regime 2×2 reported in Table 2 of the main paper use seeds $3072/1234/42$: the from-scratch cells were replicated directly, and the finetune cells are read off the three-seed scan of Table 1, which is the same regime. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in [0.7, 1.5]$, velocity $\in [1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6]\cup[1.5, 2.0]$ and velocity from $[0, 0.8]\cup[4.5, 6.0]$. Collision uses the four-column variant (two objects' masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq$1,600 trajectories per domain).

Probe protocol. Ridge regression ($\alpha{=}1$), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained "paper" initialization, and $K{=}4$ stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads $\rho{=}0.166$ for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix F). Random-encoder and pixel-statistics controls are run alongside every probe.

From-scratch 2×2, per seed. Hardest clean split, latent nMSE (uniform/collision both-OOD, parabola r/m-OOD), seeds $3072/1234/42$: uniform off $0.192/0.230/0.246$, on $0.750/0.560/0.662$; parabola off $0.343/0.420/0.666$, on $0.375/0.494/0.473$; collision off $0.538/0.513/0.454$, on $0.635/0.661/0.649$. The parabola from-scratch baseline spans 0.343–0.666 across seeds—a spread wider than the effect being measured, which is why that cell is reported as parity rather than as a gain.

B Full Injection Results

Table 1 is the numeric form of Figure 2 of the main paper, with the seed annotations spelled out. Adding labels to the label-free prior yields exactly the strict $a{=}g$ specification, so the row serves as both the correct-physics test and the labeled control. It pools four draws per domain: two independent seed-3072 runs and the seed-1234/42 top-ups. The duplicate seed-3072 draws bracket the mean (uniform 0.206 and 0.166), so their extra weight does not bias the cell.

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.136	0.122	0.479
Fixed slot	0.162°	0.142	0.625
+reweight ($w{=}30$)	0.132°	0.121°	0.614
+velocity	0.185	**0.096**	0.630
Probe	0.185	0.137°	0.657
+slot+reweight	0.141°	0.099°	0.623
Kinematic (free MLP)	0.155	0.149°	0.557°
Kinematic $a{=}g$ (grounded)	0.173	0.151	0.536°
Consistency	0.143°	0.151	0.610
Label-free prior	0.169	0.120°	0.643

Table 1: Full 27-cell injection scan (latent nMSE ↓, hardest clean split). Entries and the baseline are means over seeds $3072/1234/42$. **Bold** marks the sole seed-robust gain; ° marks overlap with the baseline's seed range; unmarked cells are worse in every seed.

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB), seed-3072 val-

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ues: baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight $1/30/100$: 0.183 / 0.114 / 0.136 (single seed), with the w=30 cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008. The second parity cell is parabola r/m-OOD, which reaches parity from w=100: three-seed 0.138 ± 0.026 at w=100 and 0.106 ± 0.017 at w=300, both overlapping the baseline's three-seed range (0.122 ± 0.007, seeds spanning 0.115–0.127), so neither resolves as a gain. Kinematic head on the reweighted slot: velocity-OOD decoded position ρ=$0.991/0.987$ (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB ($+1.25$); appearance-OOD remains the weak axis (decoded ρ $0.29 \rightarrow 0.43$, vs. 0.96 without the kinematic head).

Redundancy probe details. Protocol for §4.3 of the main paper, which reports the numbers. Two models are probed: the free-rollout baseline (no injection) and the slot-pinned model (structpos, w=30). For each, a ridge probe is fit on the dimension subset alone—the full 192, the black box [2:192], the slot [0:2], or a random pair as control—with train/test trajectories split $80/20$, so decodability is measured out of sample. Values are Pearson ρ against ground-truth position on the both-OOD partition, averaged over both coordinates, from frame embeddings; the same table holds on rolled-out latents (Appendix D). The slot absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$) without displacing the black box's redundant copy.

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot's effect is systematic in what it encodes. Table 2 uses the hardest clean split and legacy seed 3072; Table 1 reports the three-seed baselines 0.136/0.122/0.479.

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	$+0.075$	-0.064	$+0.025$

Table 2: Effect of encoded slot content by domain (latent nMSE ↓; seed 3072).

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity ($+0.075/-0.064/+0.025$) tracks velocity's role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, single-seed FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). As Table 3 shows, the slot, consistency, and probe variants degrade the mass scenarios by 3.6–11.9× in every seed while leaving the deformable failure in place.

C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.5 of the main paper) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds (∼130 runs). Free-rollout baselines (three-seed): uniform both-OOD 0.427 ± 0.039, parabola r/m-OOD 0.225 ± 0.019, collision both-OOD 0.479 ± 0.009; Physion++ horizon-64 0.258 ± 0.005 vs. teacher forcing 1.030 ± 0.028.

Teacher forcing vs. free rollout, per domain. Values behind Figure 5 of the main paper (hardest clean split, latent nMSE ↓, three-seed mean±sample std). LeWM: uniform $0.300\pm0.007 \rightarrow 0.136\pm0.008$ (2.2×), parabola $0.443\pm0.048 \rightarrow 0.122\pm0.007$ (3.6×), collision $1.152\pm0.048 \rightarrow 0.479\pm0.079$ (2.4×), Physion++ horizon-64 $1.174\pm0.041 \rightarrow 0.141\pm0.018$ (8.3×). Every teacher-forcing interval is disjoint from its free-rollout counterpart. Frozen DINOv2 over the same four settings: uniform $0.594\pm0.064 \rightarrow 0.427\pm0.039$ (1.4×), parabola $0.380\pm0.015 \rightarrow 0.225\pm0.019$ (1.7×), collision $0.803\pm0.034 \rightarrow 0.479\pm0.009$ (1.7×), Physion++ horizon-64 $1.030\pm0.028 \rightarrow 0.258\pm0.005$ (4.0×). The lever therefore survives replacing a trainable ViT-tiny with a frozen general-purpose encoder.

Presence and redundancy. The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at ρ=0.951 (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder (0.899). The 190 non-slot dimensions alone decode at 0.951, indistinguishable from the full latent; pinning the slot at weight 300 leaves the black-box copy at 0.973. Presence and the redundant copy are properties of generic visual representations.

Inert, not harmful. Unlike the trainable-encoder regime, almost everything lands within the baseline's seed band (23 of 27 cells): the only cells that separate as worse are the dynamics-head row (1.15× parabola, 1.27× collision) and the label-free prior on collision (1.07×). With the encoder frozen, injection losses can only reshape the 384→192 adapter: the redundant copy cannot be displaced, which bounds the damage—and forecloses the benefit. Injection-variant variance also roughly doubles (±0.087 vs. baseline ±0.039 on uniform), so single-seed cells on this backbone are uninformative.

D Causal Interventions on Trained Models

Protocol and full results for Test 4 (§4.3 of the main paper). All three methods run at inference on existing checkpoints—

Scenario	FR	+slot	+cons.	+c.+acc.	+probe	+pr.slot
friction_platform°	0.220±0.231	0.160±0.059	0.171±0.120	0.134±0.058	0.212±0.041	0.208±0.025
mass_collision	**0.171±0.079**	0.616±0.112	0.725±0.100	0.726±0.212	0.666±0.077	0.864±0.112
mass_dominoes	**0.069±0.013**	0.424±0.122	0.585±0.021	0.478±0.139	0.320±0.055	0.683±0.182
mass_waterpush	**0.122±0.002**	0.440±0.073	0.591±0.064	0.780±0.609	0.510±0.048	0.479±0.170

Table 3: Physion++ direct training: per-scenario rollout nMSE (↓) at 20 epochs, three-seed mean±std (3072/1234/42). Bold marks the baseline when every injected variant is worse in every seed; ° marks the unresolved, baseline-unstable scenario. Cons. = consistency, acc. = acceleration, and pr. = probe.

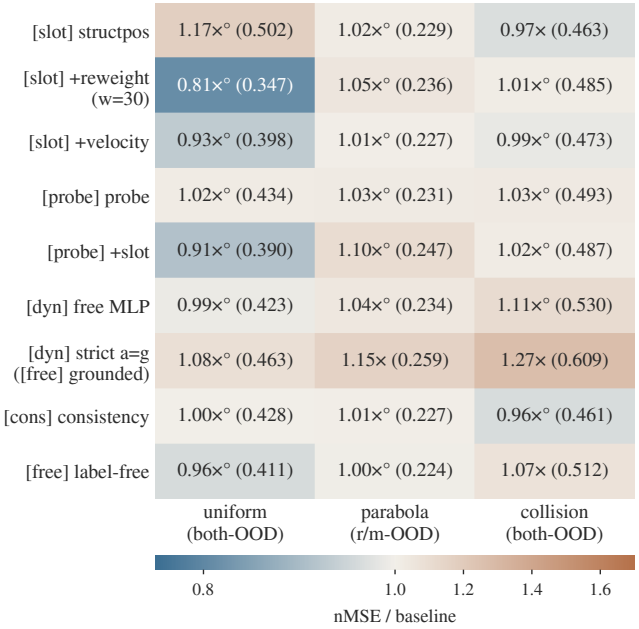


Figure 1: The injection scan repeated on the frozen-DINOv2 backbone, drawn and judged exactly as Figure 2 of the main paper (three-seed means over three-seed baseline; ° = the cell's seeds overlap the baseline's; the $a{=}g$/grounded row merged, four draws). 3 cells worse in every seed / 23 within noise / 1 below (26 of 27 fail to improve on baseline; the one cell below 1.0 separates by only a 0.001 margin at 0.97×). The dynamics-head row is the worst here too (1.27× on collision); the visually blue uniform cells are all within noise and additionally dissolve under the content-free controls of Trap 5.

no retraining—over three seeds (3072/1234/42), three domains, and both backbones. Position is read with the same ridge probe used elsewhere, fit on the training split.

Why the readout is the black box. Each intervention perturbs one channel and reads position off the *other*. Reading back the channel just written is partly trivial: the predictor copies its input forward, so the slot readout moves whether or not the perturbation reached the computation. The black-box readout is the one that reflects it (it is also what drives the pixel decoder). Predictions are recorded *before* the per-step perturbation is applied, so every number below is read off what the model itself produced.

Steering. For a target offset δ (in units of position std), we add the minimum-norm latent offset that moves the named channel's decoded position by δ—for the supervised slot, writing pos$+\delta$ directly, which is better conditioned—and report $g = \Delta(\text{decoded position})/\delta$. Table 4 reports slot gains on the black-box readout against a norm-matched random direction.

backbone	domain	g_{slot}	random control
LeWM	uniform	+1.52±0.03	−0.017
LeWM	collision	+2.01±0.20	−0.003
LeWM	parabola	+1.15±1.40	−0.017
DINOv2	uniform	+1.69±0.06	−0.011
DINOv2	parabola	+0.59±0.30	+0.001
DINOv2	collision	+1.94±0.31	−0.031

Table 4: Steering gain on the black-box readout (three-seed mean±std).

Additivity holds to within 3% ($g_{\text{slot}}+g_{\text{bb}}$ vs. steering both together), so the perturbation stays in the linear regime. *The magnitudes are not proportions*: values above 1 reflect probe $R^2{<}1$, an unperturbed action input, and the 3-frame history anchor. They support "used vs. not" and "this channel vs. that one", nothing quantitative. LeWM/parabola is excluded: its seeds read $-0.10/+0.90/+2.66$, a spread wider than the effect, and a 20× sweep of δ leaves the per-seed values essentially unchanged, so this is three different models rather than one noisy measurement.

Counterfactual patching. We replace the slot with a donor trajectory's slot at the same timestep, leaving the black box at its own values, and regress the decoded position on [own truth, donor truth]; the follow-fraction is the donor's share. The control is the same operation on a model with no injected slot, where those two dimensions carry no physical meaning: LeWM/uniform 0.015 → 0.468, DINOv2/uniform 0.112 → 0.650, DINOv2/collision 0.091 → 0.721. The LeWM parabola and collision controls are not clean (a no-slot model follows 0.42–0.51), and we exclude those cells; a collinearity explanation fits parabola (pooled own–donor correlation 0.76) but not collision (0.22), so the cause is unresolved.

Per-dimension Jacobians. $J[c,j] = \partial p_c/\partial z_j$, the sensitivity of decoded position p_c to latent coordinate z_j, averaged over 400 frames, reported per dimension so a 2-d slot and a 190-d black box share a scale. Gradient sensitivity is stan-

dard (Simonyan, Vedaldi, and Zisserman 2013; Sundararajan, Taly, and Yan 2017), here applied to an intermediate representation rather than to pixels. Unlike the interventions, it chooses no direction, which matters because a redundant code barely moves along any single one; all six cells are usable here where the black-box replacement below leaves one. Its known weakness is the reason we do not rest on it alone: a gradient is local and linear, so it reports what a small change would do rather than what changing the channel does, and some saliency methods are insensitive to the model they purport to explain (Adebayo et al. 2018). The baseline rows answer that second concern directly—the readout tracks whether the model was injected, not the input structure—and the interventions above answer the first. The baseline rows are a built-in control: with no injection those two dimensions are arbitrary and should rank mid-pack, and they do (ratio 0.80–0.98, ranks ∼90–162 of 192). After injection they rank 1st–2nd on LeWM uniform/parabola and DINOv2 uniform, at 2.3–4.0× the per-dimension sensitivity of the black box; DINOv2/parabola is the exception (ratio 0.99, ranks 111/90). The slot nonetheless holds only 0.8–4.0% of *total* sensitivity—two dimensions against 190—which is the analytic form of "both channels carry load".

Amnesic projection: a limit we quantify. We remove the linearly position-predictive subspace from the black box by iterative nullspace projection (Ravfogel et al. 2020), with a rank-matched random-ablation control. Erasing position requires removing 146–154 of the 190 dimensions, providing a stronger measure of distributed redundancy than the random-2-dimension control of Table 3 of the main paper. At this rank, however, the ablation is not discriminative: in five of six cells, removing position damages the rollout less than removing the same number of random dimensions (e.g. uniform structpos 0.608 vs. 1.116). INLP selects linearly predictive directions that also have low variance in this representation. It removes only 0.17–0.27 of the black box's variance, whereas a rank-matched random projection removes 0.77–0.82, a difference of 2.9–4.7× (mean 3.8×) across the six cells. The comparison consequently reflects removed variance as well as removed information. Searching rank-matched random subspaces for the closest variance still gives 0.74–0.77, because removing approximately 150 of 190 random directions cannot preserve four-fifths of the variance. The same pattern holds in all six cells under this control, so no comparable same-rank random control is available at the required ablation rank. Per cell (INLP / rank-matched random / variance-matched random, as fractions of the black box's total variance): uniform baseline 0.27/0.78/0.75, uniform structpos 0.17/0.82/0.77, parabola baseline 0.21/0.77/0.74, parabola structpos 0.22/0.78/0.77, collision baseline 0.25/0.81/0.77, collision structpos 0.18/0.79/0.77; INLP rank 146–154 of 190 throughout.

One further caveat we state rather than smooth over: an earlier run stopped at 24 removed dimensions showed position-removal damaging the rollout *less* than random removal—apparent support for a bypass—a contrast that disappears once INLP is run to convergence, and which the variance account explains, the mismatch being far smaller at low rank.

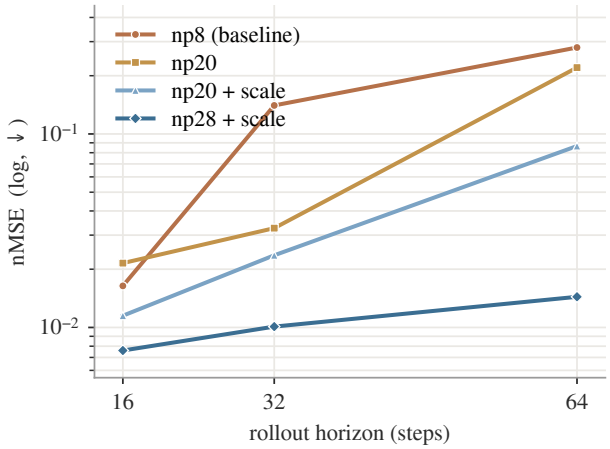


Figure 2: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H{=}8 \to 28$, +scale jitter). Long-horizon error falls monotonically to 1/19 of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

Black-box replacement. Replacing k black-box dimensions outright with a donor's (slot untouched) sidesteps the min-norm problem: on a baseline model, where the black box is the only route position can take, the min-norm patch yields a follow-fraction of 0.089 while replacing all 190 yields 0.961—an order of magnitude apart on the same model and channel. Sweeping k gives the crossing point at which moving the black box buys as much as moving the 2-d slot. A cell is usable only if its baseline both ignores a patch to the two dimensions it does not have and follows the donor once the whole black box is replaced; only LeWM/uniform passes (0.015 and 0.961), where the crossing sits near 100 dimensions. The remaining five cells fail one or both controls and their crossings are not quoted.

E Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper's main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform 0.131 → 0.068 (−48%), parabola −63% (r/m-OOD, seed 3072: 0.127 → 0.065); geometric scale jitter, only in combination with a long horizon, sets the collision record (0.172 ± 0.004 over three seeds, vs. the 0.479 ± 0.079 free-rollout baseline). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

Physion++ horizons. Values behind Figure 2, from the single-seed run of Appendix B rather than the three-seed free-rollout baseline of Appendix C; the two use different checkpoints and evaluation draws, so this ladder's $H{=}8$ entry is not the $0.141{\pm}0.018$ reported there. Overall nMSE at horizons $16/32/64$: baseline ($H{=}8$, no jitter) $0.016/0.140/0.280$; $H{=}20$ $0.022/0.033/0.220$; $H{=}16{+}$scale $0.010/0.035/0.136$; $H{=}20{+}$scale $0.012/0.024/0.087$; $H{=}28{+}$scale reaches horizon-64 nMSE 0.014. The horizon and the jitter contribute separately: at a fixed $H{=}8 \rightarrow 20$ with no jitter, horizon-64 nMSE moves only $0.280 \rightarrow 0.220$, so most of the remaining gain to 0.014 comes from adding geometric scale jitter on top of the longer horizon. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, $H{=}20$ explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062 \rightarrow 6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld's simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (see the horizon values above); appearance jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

F Metric Pathologies and Evaluation Traps

Several common metrics gave conclusions inconsistent with external prediction error. The decision rules in §4.1 of the main paper address the following five failure modes, each of which changed at least one experimental interpretation.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from $\rho{=}0.44$ to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to $100\times$ (deform_clothhit: cosine $0.610 \rightarrow 0.913$, nMSE $0.772 \rightarrow 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K{=}4$ probe ρ, $\lambda{=}50$ "won"; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes—$3{\times}10^4$–$2{\times}10^6$ at horizon ~ 28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m-OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support ($+0.10$) and Collide ($+0.07$) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pre-trained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $\rho{=}0.166$, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pre-trained model's prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Trap 5: loss re-weighting masquerades as physics. On the frozen-DINOv2 backbone, heavily re-weighting the pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from $0.427{\pm}0.039$ to $0.286{\pm}0.015$—apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches $0.330{\pm}0.013$, and weakening the pin $10\times$ while keeping the weighting reaches $0.329{\pm}0.084$—most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \rightarrow 0.115$–0.136), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision ($1.11\times$) and parabola ($1.14\times$). An apparent injection gain must be checked against a content-free control at matched loss weighting.

Auxiliary-loss dominance (measurement behind §4.3 of the main paper, Test 2). At probe weight $\lambda{=}10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder's effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \rightarrow 4$). At $\lambda{=}1$ the ratios are benign, which is the setting used in §4.2 of the main paper.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder. (v) Check any apparent injection gain against a content-free control at matched loss weighting.

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